

Chapter 28 - Magnetic Fields

Magnetic Fields

Previous goal: How an electric field can produce an electric force on a charged object.

present goal: How a magnetic field can produce a magnetic force on a moving charged particle or on a magnetic object.

Experimental fact When a charged particle moves through a magnetic field, a magnetic force $\vec{F}_B$ acts on the particle.

The Definition of $\vec{B}$

We determined the electric field $\vec{E}$ at a point by putting a test particle of charge q at rest at that point and measuring the electric force $\vec{F}_E$ acting on the particle. We then defined $\vec{E}$ as

$$\vec{E} = \frac{\vec{F}_E}{q}$$

If a magnetic monopole were available, we could define $\vec{B}$ in a similar way. Because such particles have not been found, we must define $\vec{B}$ in another way, in terms of the magnetic force $\vec{F}_B$ exerted on a moving electrically charged test particle.

Moving Charged Particle

□ In principle, we do this by firing a charged particle through the point at which $\vec{B}$ is to be defined.

→ After trials: When the particle's velocity $\vec{v}$ is along a particular axis through the point, force $\vec{F}_B$ is zero. For all other direction of $\vec{v}$, the magnitude of $\vec{F}_B$ is always proportional to $v \sin \phi$, where ϕ is the $\phi \rightarrow$ angle between the zero-force axis and the direction of $\vec{v}$. The direction of $\vec{F}_B$ is always perpendicular to the direction of $\vec{v}$.

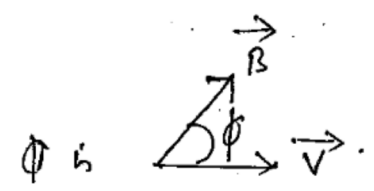
The Field

$$B = \frac{F_B}{|q|v}$$

Charge of the particle

$$\vec{F}_B = q \vec{v} \times \vec{B}$$

$$F_B = |q| v B \sin \phi$$



Finding the Magnetic force on a Particle - use thumb rule.

① The force $\vec{F}_B$ acting on a charged particle moving with velocity $\vec{v}$ through a magnetic field $\vec{B}$ is always perpendicular to $\vec{v}$ and $\vec{B}$.

Note: $\vec{F}_B$ never has a component parallel to $\vec{v}$.

This means $\vec{F}_B$ can't change the particle velocity.

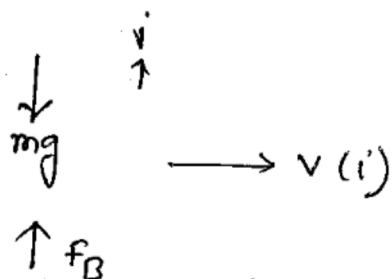
The force can only change the direction of $\vec{v}$.

Problems:

(2) A particle of mass 10g and charge 80mC moves through a uniform magnetic field, in a region where the free-fall acceleration is $-9.8\hat{j}\text{ m/s}^2$. The velocity of the particle is a constant $20\hat{i}\text{ km/s}$, which is perpendicular to the magnetic field. What, then, is the magnetic field.

$$B = \frac{F_B}{|q|v}$$

$$\begin{aligned}\vec{F}_B &= q\vec{v} \times \vec{B} \\ &= |q|vB\sin\phi.\end{aligned}$$



$\vec{B}$ must be in $-\hat{k}$ direction

$$\vec{B} = B_z \hat{k} = -\left(\frac{mg}{qv}\right) \hat{k} = (-0.061)\text{T} \hat{k}$$

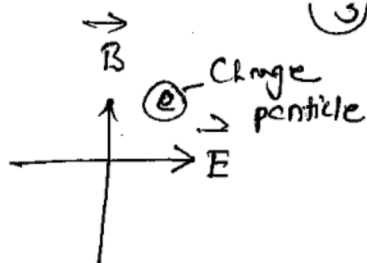
Discovery of Electron.

Canceling forces.

$$|q|E = |q|vB \sin(90^\circ) = |q|vB$$

$$\text{or } v = \frac{E}{B} \quad (\text{opposite force cancelling})$$

Thus the crossed fields allow us to measure the speed of the charged particle passing through them.

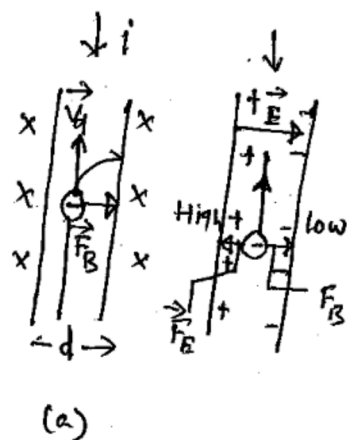


Hall Effect

We see, a beam of electrons in a vacuum can be deflected by a magnetic field. Does magnetic field deflected conduction electrons in metal?

yes $\rightarrow$ Hall effect

As time goes on, electrons move to the right, mostly piling up on the right edge of the strip.



Equilibrium. An equilibrium quickly develops in which the electric force on each electron has increased enough to match the magnetic force. The force due to $\vec{B}$ and $\vec{E}$ are in balance.

A Hall potential difference V is associated with the electric field across strip width d .

$$V = Ed.$$

Number Density When the electric and magnetic forces are in balance,

$$eE = ev_d B \rightarrow 0$$

$$v_d = \frac{J}{ne} = \frac{i}{neA} \quad (11) \quad | \quad v (= i/A) \text{ is the current density in the strip}$$

n is the number density of

$$n = \frac{B_i}{V_{le}} \quad [v = Ed] \quad \text{Charge Carriers}$$

$$d = \frac{A}{n} \quad \text{thickness of the strip}$$

Drift speed: It is also possible to use the Hall effect to measure directly the drift speed v_d of the charge carriers.

Moving Conductor: When a conductor begins to move at speed v through a magnetic field, its conduction electrons do so. (like moving conduction electrons). An electric field $\vec{E}$ and potential difference V are quickly set up.

$$eE = evB$$

$$\rightarrow \text{conductor's speed} \quad | \quad v = Ed$$

$$\boxed{V = vBd}$$

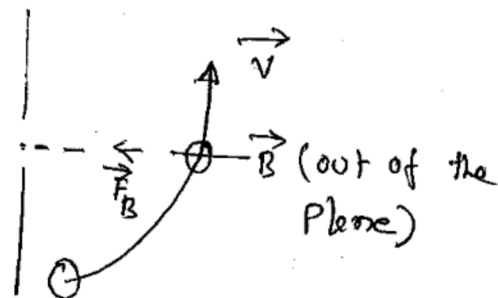
Such a motion-caused circuit

potential difference can be of serious concern in some situation. (When a conductor in an orbiting satellite moves through Earth magnetic field, - However, if a conducting line, dangles from the satellite, the potential produced along the line might be used to maneuver the satellite.

A circulating Charged Particle — Now for magnetic field

$$0 \quad \vec{F}_B = q \vec{v} \times \vec{B} \text{ continuously}$$

deflects the electrons, and because $\vec{v}$ and $\vec{B}$ are always perpendicular to each other, this deflection causes the electron to follow a circular path.



$$\vec{F}_B = q \vec{v} \times \vec{B}$$

Note: $\vec{E} = \frac{\vec{F}_E}{q}$

$$\vec{B} = \frac{\vec{F}_B}{|q|v}$$

→ From Newton's 2nd law,

$$F = m \left(\frac{v^2}{r} \right)$$

[uniform circular motion]

we have, $|q|vB = \frac{mv^2}{r} \quad \text{--- (1)}$

solving for r , we find the radius of the circular path as

$$r = \frac{mv}{|q|B} \quad (\text{radius})$$

The period T is equal to the circumference divided by the speed!

$$T = \frac{2\pi r}{v} = \frac{2\pi}{v} \frac{mv}{|q|B} = \frac{2\pi m}{|q|B} \quad (\text{period})$$

The frequency, $f = \frac{1}{T} = \frac{|q|B}{2\pi m}$ (frequency)

Angular frequency ω of the motion is then

$$\omega = 2\pi f = \frac{|q|B}{m} \quad (\text{Angular frequency})$$

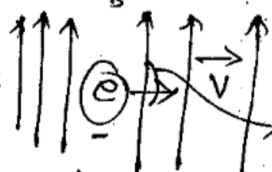
T , f , and ω don't depend on the speed of the particle!

Sample problem 28.03

Helical motion of a charged particle

in a magnetic field

$$\vec{B} (4.55 \times 10^{-4} \text{ T})$$



(kinetic energy = 22.5 eV) moves into
(65.5°)

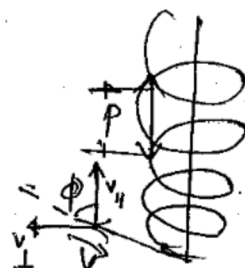
a region of uniform magnetic field $\vec{B}$ of magnitude
 $4.55 \times 10^{-4} \text{ T}$.

What is the pitch of the helical path taken
by the electron?

① → The pitch p is the distance the electron travels
parallel to the magnetic field.

$$p = v_{\parallel} T \quad (\text{one non-zero angle})$$

$$= [v \cos \phi] \frac{2\pi m}{|q| B}$$



$$v_{\parallel} = v \cos \phi$$

$$v_{\perp} = v \sin \phi$$

Calculating the electron's speed from
its kinetic energy, find that

$$v = 2.81 \times 10^6 \text{ m/s} \rightarrow$$

$$p = (2.81 \times 10^6 \text{ m/s}) (\cos 65.5^\circ) \times \frac{2\pi (9.11 \times 10^{-31} \text{ kg})}{(1.60 \times 10^{-19} \text{ C}) (4.55 \times 10^{-4} \text{ T})}$$

$$= 9.16 \text{ cm.}$$

Home work

Sample problem 28.04

Cyclotrons The key to the operation of the cyclotron is that the frequency f at which the proton circulates in the magnetic field (does not depend on its speed) must be equal to the fixed frequency f_{osc} .

$$f = f_{osc}$$

$$f = \frac{qB}{2\pi m}$$

$$qB = 2\pi m f_{osc}$$

The oscillator is designed to work at a single fixed frequency f_{osc} . To tune the cyclotron by varying B until it is satisfied.

$$\text{And } r = \frac{mv}{qB}$$

Synchrotron (i) 50 MeV — need relativistic case.

(ii) for a 500 GeV proton in a magnetic field of 1.5 T, the path radius is 1.1 km. The corresponding magnet for a conventional cyclotron of the proper size would be impossible expensive.

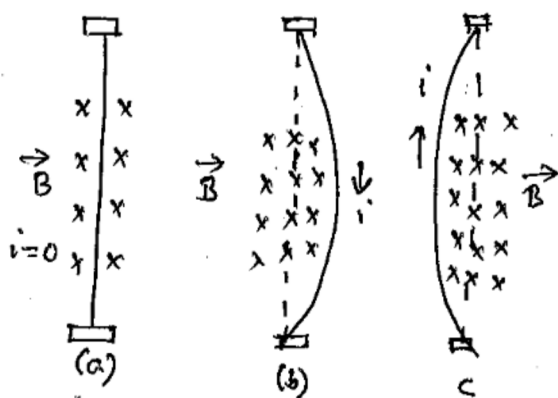
Now (i) B and f_{osc} is varying during the accelerating cycle.

(ii) the protons follow a circular path — not a spiral path. Thus, the magnet need extend only along the circular path, — not a ~~spiral~~ over $4 \times 10^6 \text{ m}^2$.

→ The circular path, however, still must be large if high energies are to be achieved.

Magnetic force on a Current-carrying wire

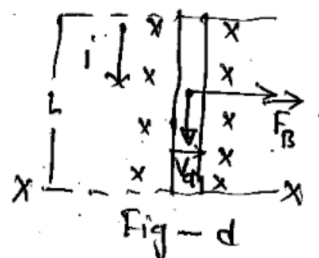
(8)



A force acts on a current through a $\vec{B}$ field.

Note: We have already seen that a magnetic field exerts a sideways force on electrons moving in a wire. This force must then be transmitted to the wire itself, because the conduction electrons can't escape sideways out of the wire.

Find the force: Consider a length L of the wire (Fig-d). All the conduction electrons in this section wire will drift in a time $t = \frac{L}{v_d}$. Thus, in that time a charge given by,



$q = it = i \frac{L}{v_d}$ will pass through $x-x$ plane.

Now, $F_B = q v_d B \sin \phi = \frac{iL}{v_d} v_d B \sin 90^\circ = iLB$ [$\vec{B}$ is perpendicular to wire]

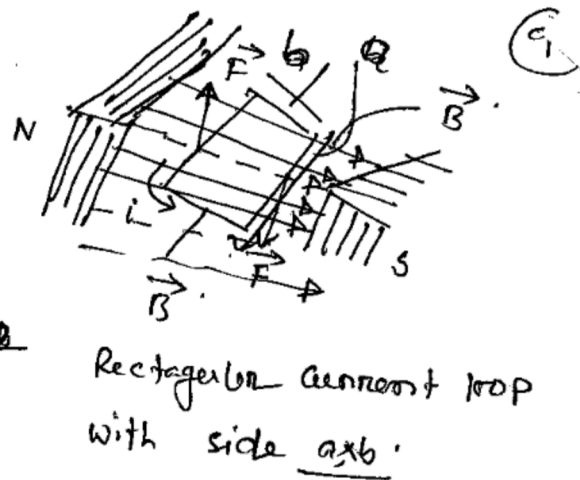
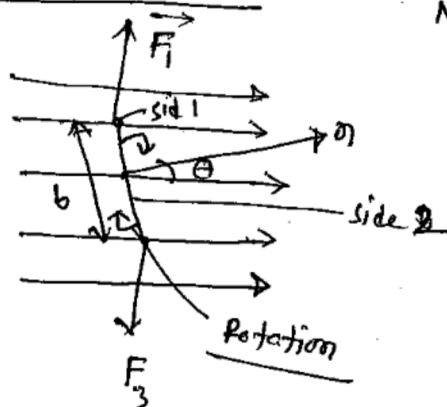
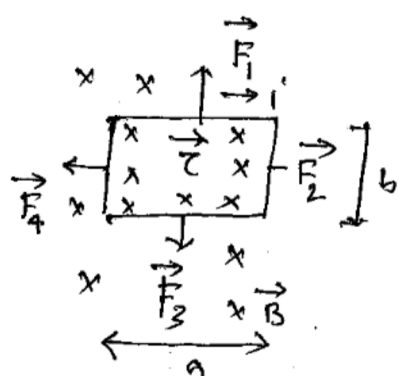
If the magnetic field is not perpendicular to the wire,

$$\vec{F}_B = i \vec{L} \times \vec{B}$$

$$F_B = iLB \sin \phi. \quad [\phi \text{ angle between } \vec{L} \text{ and } \vec{B}]$$

Curved wire: If a wire is not straight or the field is not uniform, we can imagine the wire broken up into small straight segments. The force on the wire as a whole is then the vector sum of all the forces on the segments that make it up. Then $d\vec{F}_B = i d\vec{L} \times \vec{B}$ — (*)

Torque on a Current loop



The net force on the loop is the vector sum of the forces acting on its four sides.

$$F_2 = i b B \sin(90^\circ - \theta) = i b B \cos \theta$$

F_4 has the same magnitude of F_2 but the opposite direction, and cancel out.

The situation is different for sides 1 and 3. For them $\vec{L}$ is perpendicular to $\vec{B}$, so the forces $\vec{F}_1$ and $\vec{F}_3$ have the common magnitude $i a B$. Because these two forces have opposite directions, they do not tend to move the loop up or down. However, these two forces do not share the same line of action so they produce a net torque.

Ⓜ The torque tends to rotate the loop so as to align its normal vector $\vec{n}$ with the direction of the magnetic field $\vec{B}$. That torque has moment arm $(b/2) \sin \theta$ about the central axis of the loop.

$$\tau' = (i a B \frac{b}{2} \sin \theta) + (i a B \frac{b}{2} \sin \theta) = i a b B \sin \theta$$

for coil of N turns $\rightarrow \tau = N \tau' = (N i A) B \sin \theta$.

for circular coil: $\rightarrow \tau = (N i \pi r^2) B \sin \theta$.

Magnetic Dipole Moment

(10)

A torque acts to rotate a current-carrying coil placed in a magnetic field. In that sense, the coil behaves like a bar magnet placed in the magnetic field. Thus, like a bar magnet, a current-carrying coil is said to be a magnetic dipole. Moreover, to account for the torque on the coil due to the magnetic field, we assign a 'magnetic dipole moment' μ to the coil.

$$\mu = N i A \quad (\text{magnetic moment})$$

$\downarrow$ number of turns of the coil, i is through the coil. A is the area enclosed by each turn of the coil. $1 \mu = A \cdot m^2$.

Torque:

$$\tau = \mu B \sin \theta \quad | \quad \theta \text{ is angle between } \vec{\mu} \text{ and } \vec{B}$$

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad | \quad \text{Analogy } \vec{\tau} = \vec{p} \times \vec{E}$$

electric field case.

Energy:

$$U(\theta) = -\vec{p} \cdot \vec{E}$$

In strict analogy, we can write for the magnetic case.

$$U(\theta) = -\vec{\mu} \cdot \vec{B}$$

In each case the energy due to field is equal to the negative of the scalar product of the corresponding dipole moment and the field vector.

Magnetic dipole lowest energy: $-\mu B \cos 0 = -\mu B$ (Line up with magnetic field)
It has highest energy $-\mu B \cos 180^\circ = +\mu B$. When $\vec{\mu}$ is directed opposite the field.

Work! $W = U_f - U_i$